

Exploring the World of 3D Printing

A. Warming Up

1. What do you know about 3D printing?

3D 프린팅은 3차원 물체를 실제 크기로 출력하는 기술로, 플라스틱, 금속, 종이 등 다양한 재료를 활용하여
복잡한 형상의 물체를 쉽게 제작할 수 있는 기술

2. What are some of the things you would like to create with a 3D printer?

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3. What are some of the challenges associated with 3D printing?

프린팅 시간, 재료 비용, 프린터의 크기, 프린터의 해상도 등

B. Key Expressions

1. **In contrast to** traditional printers, 3D printers are capable of producing three-dimensional objects using various materials.

2. **Unlike** regular printers that can only create flat images or documents, 3D printers can build tangible objects layer by layer.

(A) Fused deposition modeling (FDM)

(B) Stereolithography

(C) Selective laser sintering (SLS)

(D) Binder jetting

4. What advantage does 3D printing offer to designers?

- (A) Rapid transformation of concepts into tangible 3D models or prototypes
(B) Reduction of inventory management issues
(C) Minimization of waste compared to subtractive techniques
(D) Improved print quality and resolution compared to traditional printers

5. How does 3D printing contribute to reducing waste?

- (A) By using less raw material compared to subtractive techniques
(B) By eliminating the need for inventory management
(C) By allowing remote locations to fabricate objects
(D) By reducing energy consumption during the printing process

6. What potential does 3D printing have in transforming manufacturing processes?

- (A) It allows consumers to download files and create complex 3D objects within their own homes.
(B) It enables the creation of objects using different materials.
(C) It facilitates quick design iterations.
(D) It has the potential to streamline and optimize traditional manufacturing processes, making them more efficient and cost-effective.

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D. Comprehension Checkup

Choose the best answer to the following questions.

1. What is the main difference between traditional printers and 3D printers?

- (A) Traditional printers can only create flat images, while 3D printers can build three-dimensional objects layer by layer.
(B) Traditional printers are more affordable than 3D printers.
(C) Traditional printers use computer-aided design (CAD) programs, while 3D printers use computer-aided manufacturing (CAM) programs.
(D) Traditional printers require ink cartridges, while 3D printers use filament cartridges.

2. What is the core process of 3D printing?

- (A) Material is cut or drilled away to form a physical object.
(B) Material is added layer by layer to form a physical object.
(C) Material is sprayed onto a powder bed to form a physical object.
(D) Material is molded using a die or mold to form a physical object.

3. Which technique uses a heated extrusion nozzle to melt and deposit a thermoplastic filament layer by layer?

E. Grammar Checkup

Choose one that can best complete the following sentences.

1. In contrast to traditional printers, 3D printers are capable _____ three-dimensional objects using various materials.

- (A) produced (B) producing
(C) of producing (D) to produce

2. 3D printers utilize various materials and techniques _____ digital files, created using computer-aided design (CAD) or computer-aided manufacturing (CAM) programs, into real-world objects.

- (A) transform (B) were transformed
(C) to be transformed (D) to transform

3. 3D printing can be considered a form of printing, although it _____ traditional definitions.

- (A) differs from (B) differs to
(C) is differ from (D) is different to

4. One commonly used technique is fused deposition modeling (FDM), _____ a heated extrusion nozzle melts and deposits a thermoplastic filament layer by layer.

- (A) who (B) what
(C) when (D) where

5. The first 3D printer _____ in the mid-1980s by Charles W. Hull, who used

the stereolithography technique.

- (A) invented (B) inventing
(C) was developed (D) was developing

6. This technology has the potential to revolutionize manufacturing. _____ consumers to download files and create complex 3D objects, including electronics, within their own homes.
(A) allow (B) allows
(C) allowing (D) allowed

F. Vocabulary Checkup

Choose one that can best complete the following sentences.

1. This technology has gained _____ and is now available at major retailers like Staples, Best Buy, and Home Depot.
(A) expansion (B) manufacturing
(C) popularity (D) resignation
2. While 3D printers were initially used in professional settings, they are now becoming more _____ in homes, schools, and recreational spaces.
(A) common (B) digital
(C) technical (D) utilizing
3. This additive process differs from _____ processes where material is cut or

drilled away.

- (A) bypass (B) evolving
(C) manageable (D) subtractive

4. While these definitions don't exactly align with 3D printing, the technology is a(n) _____ of traditional printing methods.

- (A) bio-mechanism (B) evolution
(C) fabricate (D) objection

5. Initially an expensive commercial method, recent years have seen the _____ of more affordable desktop professional and consumer 3D printers.

- (A) design (B) emergence
(C) plasticity (D) urgency

6. It allows designers to rapidly _____ concepts into tangible 3D models or prototypes, facilitating quick design iterations.

- (A) inventory (B) manage
(C) locate (D) transform

G. Guided Composition

Using the useful expressions given, put the following Korean into English.

1. 당신은 Amazon.com과 같은 온라인 플랫폼을 통해 3D 프린터를 구매할 수 있다.
(purchase)